

Experiment/Case Study No.: 4

1. Experiment Title:

Use function types, scope, and closures; apply try-catch; build a palindrome checker.

2. Software/Tools Required:

- Visual Studio Code (or any text editor)
- Web Browser (Google Chrome/Microsoft Edge/Mozilla Firefox)

3. Experiment Program Code:

Html file:

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Palindrome Checker</title>

<link rel="stylesheet" href="style.css">

<script src="script.js"></script>

</head>

<body>

<div class="container">

<h2>&img alt="GitHub logo" data-bbox="158 681 178 691"/> Palindrome Checker</h2>

<label for="word">Enter a Word</label>

<input

type="text"

id="word"

placeholder="e.g. madam">

<input

type="button"
```

```
value="Check Word"

onclick="checkPalindrome()">

</div>

</body>

</html>
```

```
Javascript file:
// Function to validate registration number

function validateRegistration(regNo) {

// Rule 1

if (regNo === "") {

throw "Registration number should not be empty.";

}

// Rule 2

if (regNo.length !== 10) {

throw "Registration number should be exactly 10 characters.";

}

// Rule 3

if (!/^[A-Z]{2}/.test(regNo)) {

throw "First two characters must be uppercase alphabets (State Code).";

}

// Rule 4

if (!/^[A-Z]{2}[0-9]{2}/.test(regNo)) {

throw "Characters 3 and 4 must be digits (District Code).";

}

// Rule 5

if (!/^[A-Z]{2}[0-9]{2}[A-Z]{2}/.test(regNo)) {

throw "Characters 5 and 6 must be uppercase alphabets (Series).";

}

// Rule 6
```

```

if (!/^[A-Z]{2}[0-9]{2}[A-Z]{2}[0-9]{4}$/.test(regNo)) {

    throw "Last four characters must be digits (Vehicle Number).";

}

return true;

}

// Function called when button is clicked

function checkRegistration() {

    var regNo = document.getElementById("regNo").value.trim();

    var result = document.getElementById("result");

    try {

        validateRegistration(regNo);

        alert("✅ Vehicle Registration Successful!");

        result.innerHTML = `

<div class="success">

<h3>✅ Registration Successful</h3>

<hr>

<p><strong>Registration Number :</strong> ${regNo}</p>

<p><strong>Status :</strong> Valid Vehicle Registration Number</p>

</div>

`;

        // Clear textbox

        document.getElementById("regNo").value = "";

        // Focus again

        document.getElementById("regNo").focus();

    }

    catch(error) {

        alert("❌ " + error);

        result.innerHTML = `

<div class="error">

```

```
<h3>❌ Invalid Registration Number</h3>
```

```
<hr>
```

```
<p><strong>Entered Number :</strong> ${regNo || "None"}</p>
```

```
<p><strong>Reason :</strong> ${error}</p>
```

```
</div>
```

```
`;
```

```
window.onload = function () {
```

```
document.getElementById("regNo").focus();
```

```
};
```

```
}
```

```
}
```

```
style.css file:
```

```
{
```

```
margin:0;
```

```
padding:0;
```

```
box-sizing:border-box;
```

```
font-family: Segoe UI, Tahoma, Geneva, Verdana, sans-serif;
```

```
}
```

```
body{
```

```
margin:0;
```

```
height:100vh;
```

```
background-image:
```

```
linear-gradient(rgba(0,0,0,.4), rgba(0,0,0,.4)),
```

```
url("bg.jpg");
```

```
background-size:cover;
```

```
background-position:center;
```

```
background-repeat:no-repeat;
```

```
display:flex;
```

```
justify-content:center;

align-items:center;

}

.container{

width:430px;

padding:35px;

background:rgba(255,255,255,.18);

backdrop-filter:blur(12px);

-webkit-backdrop-filter:blur(12px);

border-radius:18px;

border:1px solid rgba(255,255,255,.3);

box-shadow:0 10px 35px rgba(0,0,0,.35);

text-align:center;

}

h2{

color:white;

margin-bottom:25px;

font-size:30px;

}

label{

display:block;

text-align:left;

color:white;

font-size:18px;

margin-bottom:10px;

font-weight:bold;

}

input[type=text]{

width:100%;
```

```
padding:14px;

border:none;

border-radius:10px;

font-size:16px;

outline:none;

margin-bottom:20px;

}

input[type=text]:focus{

box-shadow:0 0 10px #00d4ff;

}

input[type=button]{

width:100%;

padding:14px;

border:none;

border-radius:10px;

background:#00a8ff;

color:white;

font-size:17px;

font-weight:bold;

cursor:pointer;

transition:.3s;

}

input[type=button]:hover{

background:#0078d7;

transform:translateY(-2px);

}

.result{

color:white;

text-align:left;
```

```
font-size:22px;

line-height:45px;

margin-top:15px;

}

.success{

color:#00ff88;

font-weight:bold;

}

.fail{

color:#ff5555;

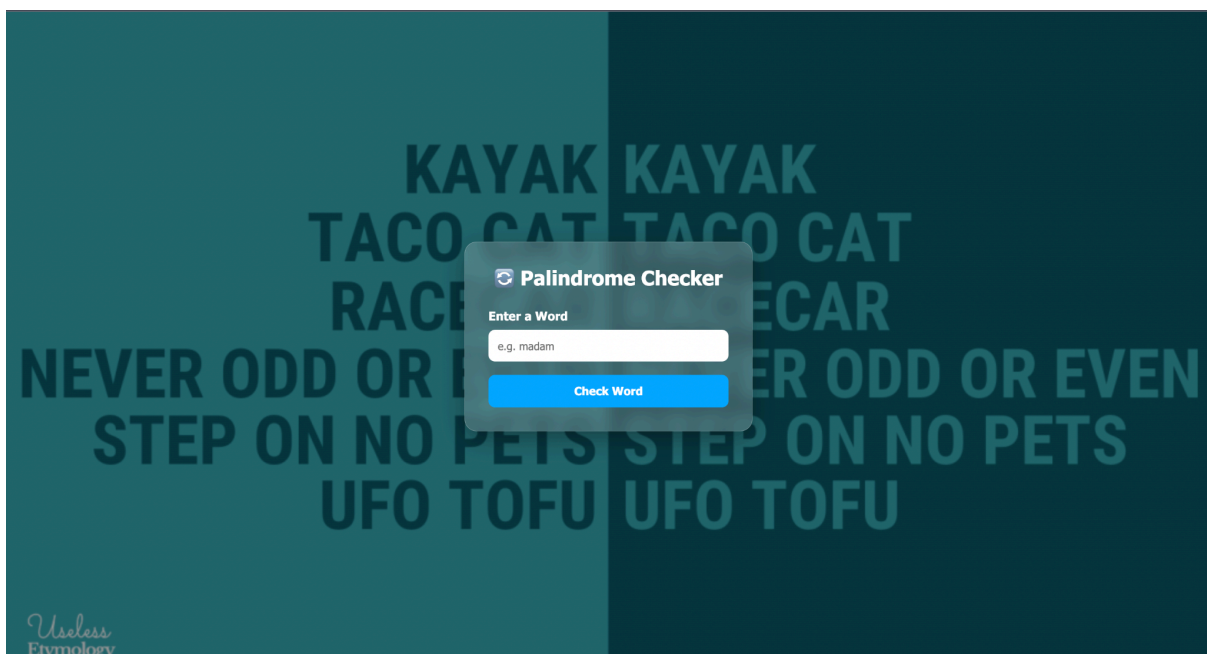
font-weight:bold;

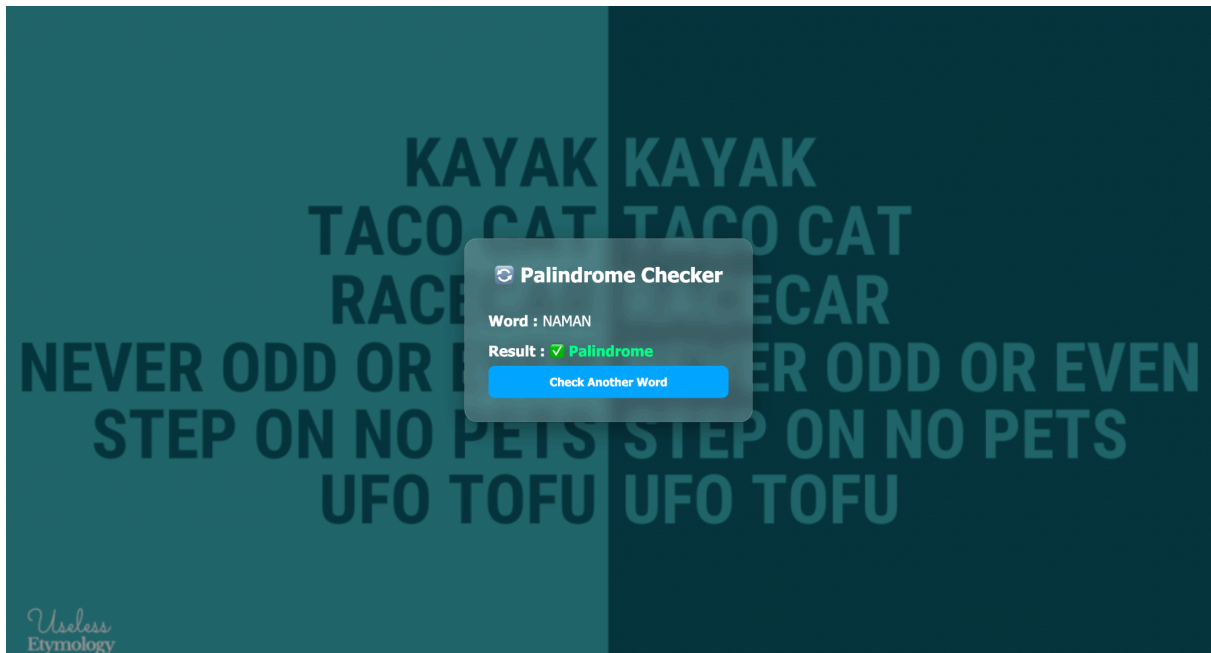
}
```

4. Output:

NAME: HONEY KALEKAR
PRN: 24070521293

SEC: A(A2)
SEM: 5TH





5. Case Study Title:

Vehicle Registration Number Checker

6. Case Study Program Code:

Html file:

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Vehicle Registration Validator</title>

<link rel="stylesheet" href="style.css">

<script src="script.js"></script>

</head>

<body>

<!-- Header -->

<header>
```

```
<div class="header-content">

<div class="logo"><img alt="Car icon" data-bbox="231 113 248 123"/></div>

<div class="header-text">

<h1>Vehicle Registration Validator</h1>

<p>Validate Indian Vehicle Registration Numbers</p>

</div>

</div>

</header>

<!-- Main Container -->

<div class="main-container">

<!-- Left Card -->

<div class="card">

<div class="card-title">

Vehicle Registration

</div>

<label for="regNo">

Registration Number

</label>

<input

type="text"

id="regNo"

maxlength="10"

placeholder="Example : MH31AB1234">

<button onclick="checkRegistration()">

<img alt="Car icon" data-bbox="121 729 138 739"/> Validate Registration

</button>

<div id="result">

</div>

</div>

<!-- Right Card -->

<div class="card">
```

`<div class="card-title">`

Registration Rules

`</div>`

`<ul>`

`<li>✔ Registration number should not be empty.</li>`

`<li>✔ Length should be exactly 10 characters.</li>`

`<li>✔ First two characters must be uppercase alphabets.</li>`

`<li>✔ Next two characters must be digits.</li>`

`<li>✔ Next two characters must be uppercase alphabets.</li>`

`<li>✔ Last four characters must be digits.</li>`

`</ul>`

`<hr>`

`<div class="card-title examples-title">`

Valid Examples

`</div>`

`<div class="examples">`

`<span>MH31AB1234</span>`

`<span>DL01CD5678</span>`

`<span>MP20XY9876</span>`

`</div>`

`</div>`

`</div>`

`<!-- Information Section -->`

`<section class="info">`

`<h2>About this Validator</h2>`

`<p>`

This application validates Indian-style vehicle registration numbers

using JavaScript functions and exception handling (try-catch).

It verifies the entered registration number according to the

specified format and displays whether it is valid or invalid,

along with an appropriate message.

</p>

</section>

</body>

</html>

Javascript file:

// Function to validate registration number

```
function validateRegistration(regNo) {
```

// Rule 1

```
if (regNo === "") {
```

```
    throw "Registration number should not be empty.";
```

```
}
```

// Rule 2

```
if (regNo.length !== 10) {
```

```
    throw "Registration number should be exactly 10 characters.";
```

```
}
```

// Rule 3

```
if (!/^[A-Z]{2}/.test(regNo)) {
```

```
    throw "First two characters must be uppercase alphabets (State Code).";
```

```
}
```

// Rule 4

```
if (!/^[A-Z]{2}[0-9]{2}/.test(regNo)) {
```

```
    throw "Characters 3 and 4 must be digits (District Code).";
```

```
}
```

// Rule 5

```
if (!/^[A-Z]{2}[0-9]{2}[A-Z]{2}/.test(regNo)) {
```

```
    throw "Characters 5 and 6 must be uppercase alphabets (Series).";
```

```
}
```

// Rule 6

```
if (!/^[A-Z]{2}[0-9]{2}[A-Z]{2}[0-9]{4}$/ .test(regNo)) {
```

```
    throw "Last four characters must be digits (Vehicle Number).";
```

```
}
```

```

return true;

}

// Function called when button is clicked

function checkRegistration() {

var regNo = document.getElementById("regNo").value.trim();

var result = document.getElementById("result");

try {

validateRegistration(regNo);

alert("✅ Vehicle Registration Successful!");

result.innerHTML = `

<div class="success">

<h3>✅ Registration Successful</h3>

<hr>

<p><strong>Registration Number :</strong> ${regNo}</p>

<p><strong>Status :</strong> Valid Vehicle Registration Number</p>

</div>

`;

// Clear textbox

document.getElementById("regNo").value = "";

// Focus again

document.getElementById("regNo").focus();

}

catch(error){

alert("❌ " + error);

result.innerHTML = `

<div class="error">

<h3>❌ Invalid Registration Number</h3>

<hr>

<p><strong>Entered Number :</strong> ${regNo || "None"}</p>

<p><strong>Reason :</strong> ${error}</p>

</div>

```

```

};

window.onload = function () {

document.getElementById("regNo").focus();

};

}

}

```

Style.css file:

```

@import url('https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;500;600;700&display=swap');

*{

margin:0;

padding:0;

box-sizing:border-box;

font-family:'Poppins',sans-serif;

}

body{

margin:0;

font-family:Poppins,sans-serif;

background:

linear-gradient(

rgba(0,0,0,.55),

rgba(0,0,0,.55)

),

url("carbg.png");

background-size:cover;

background-position:center;

background-attachment:fixed;

min-height:100vh;

}

/* Header */

header{

padding:40px 0;

```

```
}

.header-content{

width:90%;

margin:auto;

display:flex;

align-items:center;

gap:20px;

color:white;

}

.logo{

font-size:55px;

}

.header-text h1{

font-size:42px;

font-weight:700;

margin-bottom:10px;

}

.header-text p{

font-size:18px;

opacity:.9;

}

/* Navigation */

nav{

background:rgba(255,255,255,.08);

backdrop-filter:blur(12px);

border-radius:15px;

width:80%;

margin:auto;

}

nav span{

color:#0b5fa5;
```

```
font-weight:600;

cursor:default;

}

nav a:hover{

color:#ff9800;

}

/* Hero Section */

.hero{

width:92%;

margin:30px auto;

position:relative;

border-radius:18px;

overflow:hidden;

box-shadow:0 10px 25px rgba(0,0,0,.2);

}

.hero img{

width:100%;

height:350px;

object-fit:cover;

display:block;

}

.hero::before{

content:"";

position:absolute;

left:0;

top:0;

width:100%;

height:100%;

background:rgba(0,0,0,.45);

}

.hero-text{
```

```
position: absolute;

top: 50%;

left: 60px;

transform: translateY(-50%);

color: white;

}

.hero-text h2{

font-size: 42px;

margin-bottom: 15px;

}

.hero-text p{

font-size: 18px;

width: 500px;

line-height: 1.7;

}

/* Main */

.main-container{

width: 90%;

margin: 40px auto;

display: grid;

grid-template-columns: 1fr 1fr;

gap: 30px;

}

/* Cards */

.card{

background: rgba(255, 255, 255, .15);

backdrop-filter: blur(12px);

padding: 30px;

border-radius: 18px;

box-shadow: 0 12px 35px rgba(0, 0, 0, .3);

color: #fff;
```

```
}

.card-title{

background:#0b5fa5;

color:white;

padding:15px;

font-size:20px;

font-weight:600;

}

/* Left Card */

.card label{

color:white;

margin-top:18px;

display:block;

font-weight:600;

}

.card input{

width:100%;

padding:14px;

border:none;

border-radius:10px;

margin-top:8px;

background:rgba(255,255,255,.20);

color:white;

font-size:15px;

}

.card input::placeholder{

color:#ddd;

}

.card button{

width:100%;

padding:15px;
```

```
margin-top:25px;

background:#ff5a3c;

border:none;

border-radius:10px;

color:white;

font-size:16px;

font-weight:bold;

cursor:pointer;

transition:.3s;

}

.card button:hover{

background:#ff784e;

transform:translateY(-2px);

}

/* Result */

#result{

margin:20px 25px 30px;

}

/* Right Card */

/* Right Card */

.card ul{

padding:25px 40px;

}

.card ul li{

margin-bottom:14px;

line-height:1.8;

color:white;

font-size:16px;

}

.card hr{

margin:10px 25px;
```

```
border:1px solid #ddd;

}

/* Examples */

.examples-title{

background:none;

color:#0b5fa5;

padding:15px 25px 5px;

}

.examples{

display:flex;

flex-direction:column;

gap:10px;

padding:0 25px 25px;

}

.examples span{

background:rgba(255,255,255,.15);

padding:12px;

border-radius:8px;

color:white;

border-left:5px solid #00ff88;

}

/* About */

.info{

width:90%;

margin:30px auto;

padding:30px;

background:rgba(255,255,255,.15);

backdrop-filter:blur(12px);

border-radius:18px;

color:white;

}
```

```
.info h2{

color:#0b5fa5;

margin-bottom:15px;

}

.info p{

color:white;

line-height:1.8;

text-align:justify;

}

/* Success */

.success{

background:#e9f9ef;

border-left:6px solid #28a745;

padding:18px;

border-radius:10px;

color:#222;

}

.success p{

color:#222;

font-size:17px;

line-height:1.8;

}

.success strong{

color:#111;

}

.success h3{

color:#28a745;

margin-bottom:10px;

}

/* Error */

.error{
```

```
background:#ffecec;

border-left:6px solid #dc3545;

padding:18px;

border-radius:10px;

color:#222;

}

.error p{

color:#222;

}

.error strong{

color:#111;

}

.error h3{

color:#dc3545;

margin-bottom:10px;

}

/* Mobile */

@media(max-width:900px){

.main-container{

flex-direction:column;

}

header{

padding:20px;

}

.header-text h1{

font-size:26px;

}

nav{

gap:20px;

flex-wrap:wrap;

}
```

```
body::before{

width:260px;

height:180px;

}

}
```

7. Output:

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PRN: 24070521293

SEC: A(A2)
SEM: 5TH

The screenshot shows the 'Vehicle Registration Validator' web application. At the top left, there is a logo of a red car and the title 'Vehicle Registration Validator' with the subtitle 'Validate Indian Vehicle Registration Numbers'. A notification banner at the top right says '127.0.0.1:5500 says' followed by an error message: 'First two characters must be uppercase alphabets (State Code)'. Below this, the main interface is divided into two panels. The left panel, titled 'Vehicle Registration', has a text input field labeled 'Registration Number' containing 'mh3lbr2345' and a red 'Validate Registration' button. The right panel, titled 'Registration Rules', lists five rules with green checkmarks: 'Registration number should not be empty.', 'Length should be exactly 10 characters.', 'First two characters must be uppercase alphabets.', 'Next two characters must be digits.', and 'Next two characters must be uppercase alphabets.'. Below the rules is a section titled 'Valid Examples' with three input fields containing 'MH31AB1234', 'DL01CD5678', and 'MP20XY9876'.

This screenshot shows the same web application after a successful validation. The notification banner is gone. In the 'Vehicle Registration' panel, the 'Registration Number' field now contains 'Example : MH31AB1234'. The 'Validate Registration' button is now green. Below the button, a green box displays a success message: 'Registration Successful', followed by 'Registration Number : MH31BR2345' and 'Status : Valid Vehicle Registration Number'. The 'Registration Rules' and 'Valid Examples' panels remain unchanged from the previous screenshot.

8. Result/Conclusion:

The experiment and case study were completed successfully. The **Palindrome Checker** demonstrated the implementation of JavaScript function types, variable scope, closures, and exception handling using **try...catch** to determine whether a given word or sentence is a palindrome. The **Vehicle Registration Number Checker** successfully validated registration numbers against the required format using JavaScript functions, regular expressions, and input validation. Both applications effectively demonstrated the practical use of JavaScript concepts in developing reliable, interactive, and user-friendly web applications.